

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 02)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1 Describe how ozone is formed and destroyed naturally in the stratosphere.
[3 marks]

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0 1 . 2 Explain the properties of CFCs that made them useful, and the properties that cause ozone depletion (the Rowland-Molina hypothesis).
[6 marks]

Useful properties

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Properties causing ozone depletion

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0 1 . 3 Explain why increased UVB reaching the Earth's surface is harmful to living organisms. Give **three** effects.
[3 marks]

1

2

3

0 1 . 4 State what is meant by an **ozone hole**.

[1 mark]

0 1 . 5 Give **one** reason why concentration in ppm is not used to measure ozone depletion.

[1 mark]

0 1 . 6 Give **one** reason why the first evidence for ozone depletion came from ground-based measurements.

[1 mark]

15

0 2 . 1 Explain the roles of **three** named groups of soil organisms in maintaining soil fertility.

[6 marks]

0 2 . 2

Explain why a soil pH between 5.5 and 7.0 is best for fertility.

[2 marks]

0 2 . 3

Explain how soil compaction by heavy machinery reduces fertility.

[2 marks]

10

0 3 . 1

Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

A A waste is assumed to cause pollution until research shows it is unlikely to do so

B The polluter must pay for any damage caused

C Emissions should be as low as reasonably achievable

D Only pollutants known to be toxic need to be controlled

0 3 . 2

Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

0 3 . 3 Give **two** choices about emission location that reduce the severity of pollution. [2 marks]

1

2

5

0 4 Table 1 shows the mean dry mass of wheat plants grown with two fertiliser treatments (10 replicates each).

Table 1

Treatment	Mean dry mass / g	Standard deviation / g
Organic manure	14.2	3.9
Inorganic NPK fertiliser	16.1	4.1

0 4 . 1 Explain what the standard deviation values tell you about the data. [2 marks]

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0 4 . 2 Name the statistical test that should be used to decide whether the difference between the two means is significant, and justify your choice. [2 marks]

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0 4 . 3 The test gave a probability $p = 0.30$. State what this means and what the student should conclude. [2 marks]

0 4 . 4

Explain why a statistically significant correlation does not prove a causal relationship.

[2 marks]

0 4 . 5

Suggest how the student could increase the chance of detecting a real difference if one exists.

[2 marks]

10

0 5 . 1

Explain the difference between a negative and a positive feedback mechanism.

[2 marks]

0 5 . 2

Describe **two** negative feedback mechanisms that may reduce global warming.

[4 marks]

0 5 . 3

Explain how melting permafrost and melting methane hydrate act as positive feedback mechanisms.

[4 marks]

0 5 . 4

Give **three** other positive feedback mechanisms.

[3 marks]

1

2

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0 5 . 5

Explain what is meant by a climate **tipping point**.

[2 marks]

15

06 . 1

Describe **three** factors that have restricted the growth of nuclear power.

[3 marks]

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06 . 2

Explain how a plutonium 'breeder' reactor can produce more fissile fuel than it uses.

[3 marks]

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06 . 3

Give **two** advantages and **two** disadvantages of thorium reactors compared with uranium reactors.

[4 marks]

Advantages

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Disadvantages

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0 7 . 1

Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

- A Sulfur dioxide from a coal-fired power station
- B Tropospheric ozone
- C Carbon monoxide from a car exhaust
- D Nitrogen monoxide from a diesel engine

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0 7 . 2

Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

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08 . 1

Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

- A Desalinated seawater
 B River water
 C Rainwater catchment
 D Newly discovered aquifers

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08 . 2

Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

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08 . 3

Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

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09 . 1

Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

- A Aerobic decomposition produces little methane and returns nutrients to the soil
 B It produces more electricity
 C It removes heavy metals
 D It destroys asbestos fibres

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☐
☐
☐

09 . 2

Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

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09 . 3

Give **two** features of a well-managed landfill site that reduce pollution.**[2 marks]**

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2

5

10

Table 1 shows the three forms of oxygen found in the stratosphere.

Table 1

Form of oxygen	Number of atoms per molecule
Monatomic oxygen (O)	1
Diatomic oxygen (O ₂)	2
Triatomic oxygen / ozone (O ₃)	3

10 . 1

Explain how the three forms of oxygen protect life at the surface from ultraviolet radiation.

[3 marks]

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10 . 2

Give **two** other ways in which the atmosphere supports life.**[2 marks]**

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1 0 . 3 Explain why the atmosphere is thermally stratified into layers. **[1 mark]**

1 0 . 4 The troposphere and stratosphere are the layers most affected by human activities. Discuss how human activities alter the energy processes in these two layers and evaluate the difficulties of predicting the consequences. **[9 marks]**

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Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

OR

1 1 . 2

Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

[illegible]

[illegible]

END OF QUESTIONS